

COK Community App – Account Functionality Specification

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Executive Summary

The COK Community App is a comprehensive digital ecosystem designed for an off-grid community operating on an internal, cashless economy. At its core is the **K Credit** system: A treasury-backed digital currency pegged 1:1 to USD that enables seamless transactions, property management, service delivery, and community governance.

The app supports five distinct account types, each with tailored functionality:

1. **Master Admin** (sovereign system controller)
2. **Admin** (operational managers with configurable permissions)
3. **Vendor/Service Provider** (commercial participants)
4. **Resident** (full community members)
5. **Visitor** (temporary participants with upgrade potential).

All users maintain K Credit wallets with controlled fiat conversion capabilities, ensuring economic stability through a centralized treasury and approval system.

Key features include property ownership and rental tracking, barter-based service exchanges, community voting, emergency management, and a Community Investment Fund that captures a small percentage of every transaction to support infrastructure developments and app maintenance. The system is designed with governance flexibility in mind; currently operating with manual Master Admin approvals, but architected to evolve toward rules-based, semi-automated, or fully automated processes.

This specification provides a complete functional blueprint for each account type, detailing their capabilities, interactions, approval workflows, and the economic flows that bind the community together.

App Context

- **Community:** COK (off-grid community)
- **Currency:** K Credits (1 K Credit = 1 USD baseline)

- **Language:** English
 - **Economy:** Internal, cashless, backed by a treasury
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1. Master Admin Account

Master Admin – Summary

The Master Admin is the **sovereign controller** of COK's digital and economic infrastructure. They oversee the entire treasury and money supply, manage property ownership and tenancy, define system-wide services, create and manage all other accounts, and configure admin permissions. They act as the final approval authority for settlements, payroll, and vendor offerings, while also initiating community voting and deciding how the Community Investment Fund is used. The Master Admin maintains a comprehensive Payroll List for Service Providers, defining compensation structures, payment cycles, and K Credit conversion rates. All escalated issues from Admins land at the Master Admin level, making this role the ultimate decision-maker and guardian of system integrity.

1.1 Global Dashboard

The Master Admin sees a **system-wide economic and operational view**:

Treasury Balance

- Updated by Treasury Admins based on current backing assets.
- Represents total collateral supporting all K Credits.

Treasury Deposits Total (by user type)

Aggregated value of **fiat → credits** deposits by:

- Residents
- Vendors / Service Providers
- Visitors
- Admins

Treasury Settlements Total (by user type)

Aggregated value of **credits → fiat** settlements by:

- Residents
- Vendors / Service Providers
- Visitors
- Admins

Total Credits in Wallets (by user type)

Current circulating K Credits held by:

- Residents
- Vendors / Service Providers
- Visitors
- Admins

Community Investment Fund

- Balance of the fund fed by a small % of every transaction.
- Used for:
 - App maintenance
 - Infrastructure and community developments
- Only Master Admin can decide on use (possibly informed by community voting).

1.2 Property Management

Master Admin acts as the **central property registry**:

Property List

- Add, edit, delete properties.
- **Fields:**
 - Program & real-estate specifications
 - Property type:
 - Ownership / Rental / Admin
 - Commercial / Residential / Mixed

- Photos
- Documents
- Total price (for ownership)
- Installment schedule (be able to add all required installments):
 - Installment number, due date, amount, progress note (indicates if certain features are unlocked at this stage of payment)

Assign to Owner

- Link property to an account:
 - Account ID
 - Upload contract
 - Contract date
 - Payment records:
 - Installment number, amount, proof of payment
 - System shows % paid, outstanding balance remaining, progress note, next installment amount and due date.

Assign to Tenant

- Rental relationship:
 - Upload contract
 - Contract date
 - Rental cycle (daily/weekly/monthly/annual)
 - Cycle payment
 - Payment records:
 - Cycle number, cycle amount, proof of payment
 - System shows number of cycles paid, next payment amount due and due date.

1.3 Service Catalog Management

Service List

- Global catalog of community services.
 - Master Admin can add, edit, delete services.
 - Provides standardized service categories for Vendors/Admin-run services.
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1.4 Payroll List

Master Admin maintains a centralized payroll system for Service Providers (to whom Payroll applies):

Add Service Provider to Payroll

- Search from the list of Service Provider accounts
- Select Service Provider to add to Payroll List

Configure Payroll for Service Provider

For each Service Provider on the Payroll List:

- **Select Service(s):** Choose which service(s) from the Service Provider's catalog will be included in payroll compensation
- **Description:** Text box to describe the payroll arrangement, scope of work, or terms
- **Payroll Amount:** Specify the compensation amount
- **Payroll Cycle:** Select payment frequency
 - Daily
 - Weekly
 - Monthly
- **Payment Method:** Choose payment type
 - Fiat
 - K Credits
- **K Credits Conversion Rate:** (when paying in K Credits)
 - Default: 1:1 (1 K Credit = 1 USD equivalent)

- Custom: Edit to specify incentive rate (ex. 1.2:1 to provide 20% bonus for accepting K Credits) ***This is meant to encourage spending and engagement inside the community

Manage Payroll List

- View all Service Providers on payroll
 - Edit payroll configurations
 - Remove Service Providers from payroll
 - Track payroll history and payment records
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1.5 Account Lifecycle Management

Create New Account

Master Admin can create:

- Admin
- Vendor / Service Provider
- Resident
- Visitor

Fields:

- Full Name
- KYC details
- Account type

Upgrade Visitor

- Master Admin can upgrade a visitor to Resident / Vendor / Admin.
- Existing data is retained:
 - Wallet history
 - Transaction history
 - Issues raised

Suspend / Restore Account

Suspend:

- Account ID
- Effective date
- Reason category
- Note

Restore: suspended accounts when appropriate.

Suspended accounts are restricted from new economic or community actions (view only).

1.6 Approvals Center

Central approval hub for actions originating from Admins, Vendors, Residents, and Visitors:

Admin side

- Property updates (ex. construction progress) submitted by Admin.

Vendor side

- Issue Payroll
- Fiat payment conversions (settlements)
- New merch items
- New services

Resident side

- Any defined actions that require escalation (special property requests, high-value settlement requests, etc.).

Visitor side

- High-risk or high-value actions that require approval

Settlement Approvals:

- All "**Total Credits eligible for conversion**" are *technically eligible* by default, but:
 - **Master Admin must manually approve each settlement request** for now.
 - Future: rules-based, semi or fully automated approval logic can be added.

1.7 Admin Authorizations (Role/Permission Management)

For each Admin account, Master Admin can toggle functional permissions:

- Gate Access Log
- Property Updates
- Emergency Handling
- Community Updates & Notifications
- Community Posts Management
- Supermarket Management
- Services Management
- Treasury (allows treasury operations)

This makes **Treasury Admin** a specialization of Admin via the "Treasury" authorization.

1.8 Community Voting Management

From Master Admin account:

Add New Community Voting

- Headline
- Add options:
 - Each option has a description.
- Options become visible to Residents, who can vote.

Master Admin uses votes as input to guide:

- Community Investment Fund usage
 - Policy decisions
 - Infrastructure priorities
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2. Admin Account

Admin – Summary

An Admin is an **operational manager** of the community. Their exact capabilities depend on permissions set by the Master Admin, spanning gate access, property updates, emergencies, community communications, supermarket operations, and treasury tasks. Treasury Admins handle deposits and settlements after Master Admin approval, ensuring that real-world assets match the K Credits in circulation. Admins also have their own wallets and can participate in the economy, while raising issues directly to Master Admin when internal or community problems arise.

2.1 Wallet

Each Admin has a K Credit wallet:

- **Current Balance**
- **Total Credits Deposited**
 - Credits received via fiat deposits handled by Treasury.
- **Total Credits Earned**
 - From salaries, incentives, or other internal earnings.
- **Total Credits Eligible for Conversion**
 - All credits are eligible by default.
 - Admin can submit a **settlement request**:
 - Request goes to **Master Admin** for approval.
 - Once approved, **Treasury Admin** settles it and converts credits → fiat.
- **Purchases**
 - List of scanned items they paid for with credits (supermarket, vendors, etc.).
- **Settlement Requests List**
 - Request ID
 - Amount (K Credits = USD baseline)
 - Approval Status: Approved / In Progress / Declined (by Master Admin)
 - Settlement Status: Pending / Settled (executed by Treasury Admin)

- When Settled:
 - The amount is deducted from the wallet.
 - Receipt is auto-generated and proof of payment is stored.
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2.2 Updates & Notifications

List view with:

- Date
- Priority (yellow/orange/red)
- Message

Contains system alerts, emergency updates, approvals status changes, etc.

2.3 Raise Issue / Complaint

Admins can raise internal issues **to Master Admin**:

- Level of seriousness: yellow / orange / red
- Level of urgency: yellow / orange / red
- Category
- Message

Routing logic:

- Issues raised by Admins → **Master Admin only** (super-admin level).
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2.4 Gate Access Log (if authorized)

- Scan member ID:
 - View member's authorization status (allowed, restricted, etc.).
- Create:
 - Entry timestamp on arrival.
 - Exit timestamp on departure.

Supports security, attendance, and access control.

2.5 Property Updates (if authorized)

Admins submit official property progress updates:

- Select Property ID
- Update date
- Update number
- Parameters with % completion (e.g., Foundations / Structure / Finishing, etc.)
- Development status notes
- Photos

These updates then **go to Master Admin for approval** before being visible to the Resident.

2.6 Emergency Handling (if authorized)

Admin can manage emergency incidents:

Create New Issue

- Headline
- Category (drop-down)
- Seriousness (yellow/orange/red)
- Urgency (yellow/orange/red)

Add Updates

- Update number
- Timestamp
- Status color (green/yellow/orange/red)
- Status state (Unresolved / In Progress / Resolved)
- Status note

This allows the Admin to manage the lifecycle of emergencies and coordinate responses.

2.7 Community Updates (if authorized)

Admins post official updates:

- Update number
- Category (drop-down)
- Headline
- Photo
- Update message

Official communications appear in residents' and visitors' notifications feeds.

2.8 Manage Community Posts (if authorized)

Admins moderate member-generated content:

- View community posts.
- **Flag post:**
 - Visually shows a bold red exclamation mark on the post.
 - Nudges the community member to edit their content.
- **Flag classification** (drop-down):
 - E.g., inappropriate content, spam, misinformation, etc.

This is a soft moderation system focused on correction before removal.

2.9 Supermarket Management (if authorized)

Admins manage the central supermarket inventory:

Add Item

- Barcode
- Item name, number
- Category / subcategory

- Short & long description
- Photos
- Starting inventory (units)

Manage Item

- Total available units
- Total sold:
 - By date range
 - Today / This week / This month
- Procurement batches:
 - Batch number
 - Procurement date
 - Number of new items
 - Number of damaged/returned items

Remove Item

- Item number
- Reason for removal (drop-down)
- Note

All purchases at the supermarket are paid in K Credits.

2.10 Treasury (Treasury Admin only)

Admins with "Treasury" authorization handle cash-in and cash-out:

Update Daily Treasury Amount

- Reflects current backing assets.

Deposit Credits (New Deposit)

- Member ID
- Deposit amount (fiat → credits)

- Deposit date and timestamp
- Payment method
- Upload receipt / proof of payment
- Credits issued (system-generated; typically 1:1 with USD)

Settlement Requests (Vendor/Service Provider, Resident, Visitor)

- View requests that have been **Approved by Master Admin**.
- For each:
 - Request ID
 - Amount (K Credits = USD baseline)
 - Approval Status (read-only, must be "Approved")
 - Settlement Status: Pending / Complete
- Treasury Admin marks as **Complete** after payment.
 - Wallet is automatically debited.
 - Receipt and proof of payment are generated and stored.

2.11 Payroll (Vendor Interaction & Approval)

When vendors issue payroll:

- Payroll can be:
 - Paid in **fiat**, or
 - Issued as **K Credits**:
 - With an incentive ratio (e.g., 1.2:1) – employees receive more value in credits than in fiat.
 - Vendor's **Issue Payroll** requests requiring treasury interaction or system-wide effect appear in Master Admin's **Pending Approvals**.
 - Once approved, the system pays staff in credits (or coordinates fiat via treasury) accordingly.
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3. Vendor / Service Provider Account

Vendor / Service Provider – Summary

Vendors and Service Providers are **commercial participants** in COK's cashless economy. They maintain their own merchandise and service catalogs, track inventory and sales, and receive payments in K Credits. They can request settlements to convert credits into fiat, subject to Master Admin approval and Treasury Admin execution. Service Providers may be included in the Master Admin's Payroll List, receiving regular compensation in fiat or K Credits (with optional incentive rates) for designated services. Vendors can raise issues that surface at both Admin and Master Admin levels, ensuring their operational concerns are addressed promptly.

3.1 Wallet

Vendor wallets mirror the others' structure:

- **Current Balance**
- **Total Credits Deposited**
 - Fiat deposits converted into credits for operations.
- **Total Credits Earned**
 - From sales to residents, visitors, and other users.
- **Total Credits Eligible for Conversion**
 - All credits technically eligible.
 - Vendor submits settlement requests:
 - Master Admin must approve.
 - Treasury Admin then completes settlement (credits → fiat).
- **Purchases**
 - Vendor purchases within the ecosystem (supplies, supermarket, services).
- **Settlement Requests**
 - Request ID
 - Amount

- Approval Status: Approved / In Progress / Declined (by Master Admin)
 - Settlement Status: Pending / Settled (by Treasury Admin)
 - Settlement triggers wallet deduction and archived receipts.
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3.2 Payroll Details (if applicable)

If the Service Provider is included in the Master Admin's Payroll List, they can view:

Payroll Information

- **Service(s) Included:** List of services covered under payroll
- **Description:** Payroll arrangement details and scope of work
- **Payroll Amount:** Compensation amount per cycle
- **Payroll Cycle:** Payment frequency (daily/weekly/monthly)
- **Payment Method:** Fiat or K Credits
- **K Credits Conversion Rate:** If paid in K Credits (e.g., 1:1 or 1.2:1)

Payroll History

- Payment date
- Cycle number
- Amount paid
- Payment method (Fiat or K Credits)
- Conversion rate applied
- Payment status (Pending / Completed)
- Receipt/proof of payment

Next Payment Due

- Upcoming payment date
 - Amount due
 - Cycle period covered
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3.3 Merch Management

For vendors selling physical or digital items:

Add Item

- Barcode
- Item name, number
- Category / subcategory
- Short & long descriptions
- Photos
- Starting inventory (units)

Manage Item

- Total available units
- Total sold:
 - By date range
 - Today / This week / This month
- Procurement batches:
 - Batch number
 - Procurement date
 - New items
 - Damaged/returned items

Remove Item

- Item number
- Reason for removal (drop-down)
- Note

Approval:

New merch items and potentially certain changes flow into Master Admin's **Pending Approvals** before going live.

3.4 Service Management

For services (e.g., transportation, wellness, workshops):

Add Service

- Barcode
- Service name, number
- Category / subcategory
- Service duration
- Short & long description
- Photos
- Daily capacity (max bookings per day)
- Calendly link (for scheduling)

Manage Service

- Total available units/slots
- Total sold:
 - By date range
 - Today / This week / This month

Remove Service

- Service number
- Reason for removal (drop-down)
- Note

Approval:

New services are reviewed and approved by Master Admin before being part of the official catalog.

3.5 Updates & Notifications

Vendors receive:

- Date

- Priority (yellow/orange/red)
- Message

Such as payroll approvals, payroll payments, service approvals, system updates, and emergency alerts.

3.6 Raise Issue / Complaint

Vendors can raise issues that go **to Admin and Master Admin levels**:

- Seriousness: yellow / orange / red
- Urgency: yellow / orange / red
- Category
- Message

Routing logic:

- Vendor issues → visible to **Admins and Master Admin**.
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4. Resident Account

Resident – Summary

Residents are **full-fledged members** of COK's economic, social, and governance structure. They hold K Credit wallets, manage and track their property ownership or tenancy, and can participate in both cashless and barter economies. Through the service center they request maintenance, transportation, and other services, and they help shape the community via posts and voting. Residents can raise issues that are visible to both Admins and Master Admin, ensuring their voice is heard at operational and strategic levels. Their profile includes personal information, property details with payment tracking, and the ability to offer barter services to fellow community members.

4.1 Profile & Wallet

Profile

- Upload profile photo.

- Personal introduction.

Wallet

- Current balance.
- Total Credits deposited (fiat → credits).
- Total Credits earned (e.g., jobs, barter with credits, incentives).
- Total Credits eligible for conversion:
 - All credits are eligible.
 - Resident can submit settlement requests.
 - Master Admin must approve; Treasury Admin executes.
- Purchases:
 - List of scanned items from vendors, supermarket, etc.
- Settlement requests:
 - Request ID
 - Amount
 - Approval Status (Master Admin)
 - Settlement Status (Treasury Admin)
 - Receipts and proofs saved upon settlement.

4.2 Residency & Property Details

Each resident can view structured property information:

- Full Name
- Member ID
- Member since
- **Property Details:**
 - Property Type (Commercial / Residential / Mixed)
 - Photos

- % Ownership
- Rental property: yes/no
- Contract date
- Total price
- Program details / property specs
- Documents:
 - Contracts
 - Title Deeds
- **Installment schedule:**
 - Installment number, date, amount
- **Payment records (ownership):**
 - Installment number
 - Installment amount
 - Proof of payment
 - System calculates % paid.
- **Rental cycle (if tenant):**
 - Daily/weekly/monthly/annual
 - Cycle payment
 - Payment records:
 - Cycle number
 - Cycle amount
 - Proof of payment
 - **of cycles paid.**

Updates to this information originate from Admins and must be approved by Master Admin before residents see them.

4.3 Barter Services

Residents can participate in **barter-based exchanges**:

Checkboxes for services they are willing to offer:

- Cooking & homemade food
- Housekeeping
- Babysitting
- Gardening
- Home maintenance
- Home decoration
- Knitting & crafts
- Home schooling
- Petcare
- Beauty services
- Personal products

Additional:

- Paragraph describing their offering.
- Up to 6 photos with mini descriptions.

Barter transaction logic:

A barter interaction **starts with a request**:

- One member requests a service under "barter services".
- They make an **offer** in return:
 - Either:
 - Their own service (with clear scope or time specification), or
 - K Credits (monetary component).
- Scope/time for offered services must be specified for clarity (e.g., "3 hours of gardening").

- This enables:
 - Pure barter (service for service), or
 - Hybrid barter (service + K Credits).
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4.4 Request Service – via Services Center

Residents can request services centrally:

Categories:

- Maintenance
- Supermarket
- Transportation
- Shipping
- Barter services

Requests route to:

- Relevant Vendors / Service Providers, or
- Admin-run services (depending on configuration).

Payments may be in K Credits and/or barter-based, as defined by the service and agreement.

4.5 Raise Issue / Complaint

Residents can raise issues that go **to Admin and Master Admin levels:**

- Seriousness: yellow / orange / red
- Urgency: yellow / orange / red
- Category
- Message

Routing logic:

- Resident issues → visible to **Admins and Master Admin.**

4.6 Updates & Notifications

Residents receive:

- Date
- Priority (yellow/orange/red)
- Message

Includes:

- Official community updates from Admins.
- Emergency alerts.
- Account and transaction notifications.
- Voting invitations.

4.7 Community Post

Residents can:

- Create and manage posts in the community feed.
- Posts may be flagged by Admins:
 - A flagged post shows a red exclamation mark.
 - Resident is nudged to edit it, based on flag classification.

4.8 Community Voting

Residents participate in governance via voting:

- Master Admin publishes community votes:
 - Headline.
 - Multiple options with descriptions.
- Residents can cast votes on each option.
- Results inform Master Admin decisions:

- Use of Community Investment Fund.
 - Policy and development priorities.
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5. Visitor Account

Visitor – Summary

Visitors are **temporary participants** who can still fully use K Credits for purchases and settlements while in COK. They receive relevant notifications and can submit complaints that reach both Admin and Master Admin. If they become more permanently involved in the community, their accounts can be upgraded without losing any history, providing a seamless transition from guest to resident or vendor. Visitors maintain full wallet functionality, can make purchases throughout the community, and request settlements subject to the same approval processes as other users.

5.1 Wallet

Visitors also have full wallet capability:

- Current balance.
- Total Credits deposited.
- Total Credits earned (e.g., refunds, incentives).
- Total Credits eligible for conversion:
 - All credits technically eligible.
 - Visitor can request settlement.
 - Master Admin approves; Treasury Admin settles.
- Purchases:
 - List of items/services purchased in K Credits.
- Settlement requests:
 - Request ID
 - Amount
 - Approval Status (Master Admin)

- Settlement Status (Treasury Admin)
 - Receipts and proofs stored post-settlement.
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5.2 Upgradable to Other Roles

Visitors can be **upgraded** by the Master Admin to:

- Resident
- Vendor / Service Provider
- Admin (rare, if needed)

On upgrade:

- All wallet history, transaction data, and issues raised are preserved.
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5.3 Updates & Notifications

Visitors receive:

- Date
- Priority
- Message

They see relevant announcements, emergency alerts, transaction notifications, and any specifics related to their stay/usage.

5.4 Raise Issue / Complaint

Visitors can raise issues to **Admin and Master Admin**:

- Seriousness: yellow / orange / red
- Urgency: yellow / orange / red
- Category
- Message

Routing logic:

- Visitor issues → visible to **Admins and Master Admin**.
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6. Cross-Role Interactions & Flows (High-Level)

Deposits (fiat → K Credits)

- Any user (Admin, Vendor, Resident, Visitor) can deposit fiat to receive K Credits.
- Treasury Admin processes deposits; Master Admin oversees via dashboard.

Purchases & Transfers

- Residents, Vendors, Visitors, and Admins transact using K Credits.
- A small % of every transaction is siphoned into the **Community Investment Fund**.
- Purchases create detailed line items in each user's wallet history.

Settlements (K Credits → fiat)

- Users submit settlement requests from their wallets.
- All credits are eligible by default, but:
 - **Master Admin** must approve each request.
 - **Treasury Admin** executes settlement and records proof of payment.

Payroll

- Master Admin maintains a Payroll List for Service Providers.
- For each Service Provider:
 - Selects specific services to be covered
 - Defines compensation amount, cycle, and payment method
 - Can set K Credit conversion rates (e.g., 1.2:1 for incentives)
- Service Providers view their payroll details and payment history in their accounts.
- Payroll payments are processed according to the defined cycle.

Barter

- Resident-to-resident (potentially also vendor-to-resident) barter:
 - Requester specifies:

- Required service under barter.
- Offered return: service (with clear scope/time) or credits.
- Can be pure barter or hybrid (service + credits).

Issues & Escalations

- Issues from **Residents, Vendors, Visitors**:
 - Routed to **Admins and Master Admin**.
- Issues from **Admins**:
 - Routed only to **Master Admin**.
- All issues include seriousness and urgency levels for prioritization.

Governance & Voting

- Master Admin creates votes (options + descriptions).
- Residents vote on presented options.
- Results guide Master Admin decisions, especially on Community Investment Fund usage and infrastructure priorities.

Permissions & Governance Evolution

- **For now:**
 - Master Admin manually approves conversions and key actions.
- **Future:**
 - System can evolve towards rules-based, semi-automated, or fully automated approvals (based on thresholds, user history, compliance rules, etc.), using the same structural flows described above.

Conclusion

This document provides a comprehensive, structured view of how each account type functions within the COK Community App, how they interact economically and operationally, and how control and governance flow from Master Admin down to Admins, Vendors, Residents, and Visitors within the COK community.

The system is designed with flexibility in mind, allowing for future automation and rule-based governance while maintaining clear accountability and transparency throughout all transactions and interactions. The addition of the centralized Payroll List enables systematic compensation management for Service Providers, with flexible payment options and incentive structures that encourage K Credit adoption.